125102072

MID SEM – II (PHIC 101, **AIML, AIDS, MV, CSE, IT, M&C, IIOT**) TIME: **3:00 PM – 3:50 PM**

B.Tech (**1st Semester**) Time Allowed: **50 min** Max. Marks: **15**

NOTE: Attempt any **Three questions**. Extra ANSWER SHEETS will **NOT** be supplied.

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(1) Describe de-Broglie hypothesis and derive the relation between de-Broglie wavelength and kinetic energy E_K of a non-relativistic particle having mass m. Estimate the kinetic energy of the electron whose de Broglie wavelength is 4×10^{-12} m. (The rest mass of an electron is $m_o = 9.1 \times 10^{-31}$ kg.) **[3 + 2]**

(2) Define scalar and vector fields with examples and relevant diagrams. For the vector field $\vec{A} = (x + 2y + \alpha z)\hat{\imath} + (\beta x - 3y - z)\hat{\jmath} + (4x + \gamma y + 2z)\hat{k}$, evaluate the value of α, β, γ such that $\vec{\nabla} \times \vec{A} = 0$. **[2 + 3]**

(3) What are the physical significances of Miller indices? Explain using relevant diagrams. Draw following Miller planes $(3\ 1\ 0)$, $(\bar{2}\ 1\ \bar{2})$ and $(\bar{1}\ \bar{2}\ \bar{3})$. **[2 + 3]**

(4) Write the expression for Fermi-Dirac distribution function and discuss the physical significance of Fermi energy E_F. Show that for $T \neq 0$, occupation probability at $E = E_F + \Delta E$ is same as non-occupation probability at $E = E_F - \Delta E$. **[2 + 3]**